

REGULAR ARTICLE

E-Commerce Into Rural: A Catalyst for Agricultural Mechanization in China

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ABSTRACT

While agricultural mechanization is vital for productivity, the “smallholder dilemma” in developing countries often creates prohibitive barriers to technology adoption. We identify rural e-commerce as a novel institutional catalyst that alleviates these constraints. Using a staggered DID framework applied to county-level panel data and household-level surveys in China, we demonstrate that the expansion of digital trade significantly enhances agricultural mechanization levels. Beyond a simple sales-channel effect, we uncover a “factor reallocation” mechanism by reducing transaction costs: e-commerce incentivizes land transfer and fosters the emergence of a “sharing economy” in agricultural machinery through cross-regional service markets. Notably, the policy acts as an equalizer, generating higher marginal benefits in administratively smaller counties and low-grain-output regions. By demonstrating how digital infrastructure can resolve the indivisibility of capital in fragmented agrarian systems, this study provides a new theoretical and empirical link between the digital economy and structural agricultural transformation.

JEL Classification: O13, O18, Q15

1 | Introduction

Agricultural mechanization is a cornerstone of modern agrarian transformation. By substituting traditional labor-intensive practices with advanced capital inputs—such as tractors and automated harvesting systems—mechanization fundamentally overhauls agricultural production functions, leading to significant productivity gains. A substantial body of literature has documented the multifaceted benefits of mechanization, including the mitigation of rural labor shortages (Byres and Byres 1981), the enhancement of rural household income (Lyne et al. 2018), and the sustained growth of agricultural total factor productivity (Chen et al. 2008; Guo and Wang 2022; Lu et al. 2022). Consequently, identifying effective catalysts for mechanization remains a central theme in development economics, particularly for nations striving to escape the low-productivity trap of small-scale farming.

The extant literature has identified several key drivers of agricultural mechanization, ranging from government-led agricultural subsidies (Mohammed et al. 2023) and rising household wealth (Benin 2015; Wang et al. 2016) to institutional factors such as land consolidation (Wang et al. 2020) and the emergence of specialized service markets (Zhang et al. 2017). Recently, the rapid penetration of digital technology into rural hinterlands has emerged as a potential paradigm-shifter. However, despite the proliferation of digital platforms, robust empirical evidence regarding the impact of rural e-commerce on agricultural production technology remains strikingly scarce. This study aims to fill this gap by leveraging China’s “E-Commerce into Rural Comprehensive Demonstration County Pilot” policy as a quasi-natural experiment.

China provides an ideal institutional laboratory for this inquiry for several reasons. First, its agricultural sector exhibits the

“Smallholder Dilemma” common to many developing economies—characterized by extreme land fragmentation and high transaction costs (Yang et al. 2013; Zhang et al. 2017). Second, China has leaped to become the world’s largest e-commerce market, with transaction volumes exceeding 50 trillion yuan. Finally, the “E-Commerce into Rural” initiative represents a state-led digital intervention model that is being actively emulated in other global south nations, including Egypt, India, and Vietnam (Couture et al. 2021; Liu et al. 2023). Therefore, evaluating China’s experience yields critical insights for the global “Digital for Development” (D4D) agenda.

Utilizing a staggered difference-in-differences (DID) framework on a comprehensive panel dataset of 2076 Chinese counties (2010–2020), we find that rural e-commerce has significantly boosted agricultural mechanization levels. This finding is robust to various econometric challenges, including parallel trend violations and the potential biases of heterogeneous treatment effects, and is further validated by micro-level household data from the China Family Panel Studies (CFPS).

We further unpack the underlying mechanisms at both macro and micro levels. Our analysis reveals that rural e-commerce functions as a “market-expansion” shock that lowers the relative cost of mechanization by reducing transaction costs. This occurs through two distinct yet complementary channels: facilitating land transfers to achieve internal economies of scale and fostering cross-regional mechanization service markets to achieve external economies of scale. Interestingly, the effect is significantly more pronounced in administratively smaller counties and regions with lower grain output. This suggests that e-commerce acts as a “leveling force,” allowing marginalized areas to bypass traditional capital constraints and integrate into modern production networks.

We make a significant contribution to the burgeoning literature on the economic consequences of e-commerce. While existing research has extensively documented its impact on market accessibility (Zhao et al. 2024), consumption inequality (Fan et al. 2018), rural entrepreneurship (Galloway et al. 2011; Mary George et al. 2016), and even environmental outcomes (Jensen 2007), there remains a conspicuous empirical gap regarding its influence on agricultural production technology. Specifically, while previous studies have predominantly focused on the “information channel” of digital platforms—emphasizing price transparency and reduced search costs (Deichmann et al. 2016; Kaila and Tarp 2019)—they often overlook the profound “market expansion” function that e-commerce facilitates. By addressing this oversight and identifying market expansion as a primary driver of technological adoption, our work provides a novel theoretical and empirical perspective for assessing digital interventions like the “E-commerce into Rural” initiative. This shift in focus from consumption-led growth to production-side transformation offers a more holistic understanding of digital dividends in agrarian economies.

Furthermore, our study contributes to the established literature on the determinants of agricultural mechanization. To date, scholarly inquiry has primarily focused on supply-side interventions and endowment-based drivers, such as agricultural

subsidies (Mohammed et al. 2023), rural income growth (Benin 2015; Wang et al. 2016), land consolidation (Wang et al. 2020), and achieving cross-regional mechanized harvesting (Zhang et al. 2017). However, the transformative role of the digital economy—and specifically e-commerce—in reshaping these traditional drivers remains largely under-explored. We bridge this gap by identifying e-commerce as a powerful, market-driven catalyst that rejuvenates agricultural productivity. By demonstrating how digital platforms can operationalize land transfers and optimize cross-regional service matching, we provide a reproducible blueprint for developing countries seeking to modernize their agrarian sectors in the digital age.

The remainder of this paper is organized as follows: Section 2 introduces the institutional background of the “E-Commerce into Rural” policy. Section 3 develops the theoretical framework. Section 4 describes the data and empirical strategy. Section 5 presents the empirical results and mechanism analysis. Section 6 discusses external validity and Section 7 concludes.

2 | E-Commerce Into Rural Comprehensive Demonstration County Pilot

In July 2014, the Chinese Ministry of Finance and Ministry of Commerce issued the “E-Commerce into the Rural Comprehensive Demonstration County Pilot”, marking the first step of e-commerce promotion in rural areas. This initiative aimed to facilitate the market cycle of industrial goods destined for rural areas and agricultural products destined for urban areas, while also paving the way for e-commerce in rural areas.

The implementation of this policy involves the submission of a program of e-commerce in rural areas by each county to the central government in accordance with the specific circumstances of that county. The central government then selects the demonstration counties according to the submission, providing them with approximately 20 million yuan of central financial support. The rural e-commerce pilots are subsequently tested by the central government on a regular basis to ascertain the development of e-commerce, organizational effectiveness, and performance.

The expansion of e-commerce into rural areas has been steadily advancing year by year. Annually, new counties are designated as e-commerce demonstration counties for pilot initiatives. As shown in Figure 1, the number of rural e-commerce pilot counties stood at 56 in 2014 and has remained above 200 since 2015.¹ During the period from 2014 to 2020, which coincides with our study period, the selection of rural e-commerce pilots emphasized the role of e-commerce in poverty alleviation. Accordingly, the government prioritized poor counties in this selection process.

The implementation of this policy has facilitated the rapid development of rural e-commerce, resulting in lowering market entry barriers for farmers. The growth of rural online sales has been exponential, with China’s rural online retail sales reaching 2.5 trillion yuan in 2023. This represents a year-on-year increase of 12.9% and is nearly 13 times higher than in 2014.² This once again proves that e-commerce has expanded the market demand for agricultural products.

Furthermore, the descriptive evidence in Figure 2 illustrates the dynamic divergence in agricultural mechanization between demonstration and non-demonstration counties.³ Prior to the policy intervention in 2014, both groups followed a relatively parallel trajectory. However, following the inception of the “E-Commerce into Rural” initiative, demonstration counties exhibited a markedly accelerated growth rate. By 2016, the mechanization level in these treatment areas surpassed that of the control group, with the gap widening significantly by 2020. This trend provides preliminary support for the hypothesis that rural e-commerce serves as a catalyst for technological adoption.

3 | Theoretical Framework and Hypothesis Development

The theoretical framework is built upon the interplay between transaction cost economics and the theory of economies of scale in capital adoption as shown in Figure 3. We posit that the

“E-Commerce into Rural” policy acts as a technology-driven shock that fundamentally alters market boundaries. This shock creates a powerful new incentive structure for farmers, who then respond by reallocating factors of production (land and capital) through two distinct market-based mechanisms to make mechanization profitable.

In traditional agricultural markets, farmers face substantial limitations imposed by high transaction costs (Shimamoto et al. 2015). Specifically, they struggle with information asymmetry (Jensen 2007), which makes it difficult to access accurate distant market prices and demand information. Furthermore, farmers encounter significant market access costs (Muto and Yamano 2009) associated with finding and reliably contracting with a broad base of potential buyers outside their immediate vicinity.

Rural e-commerce, as a digital technology, directly attacks these frictions. Through online platforms, it breaks the geographical

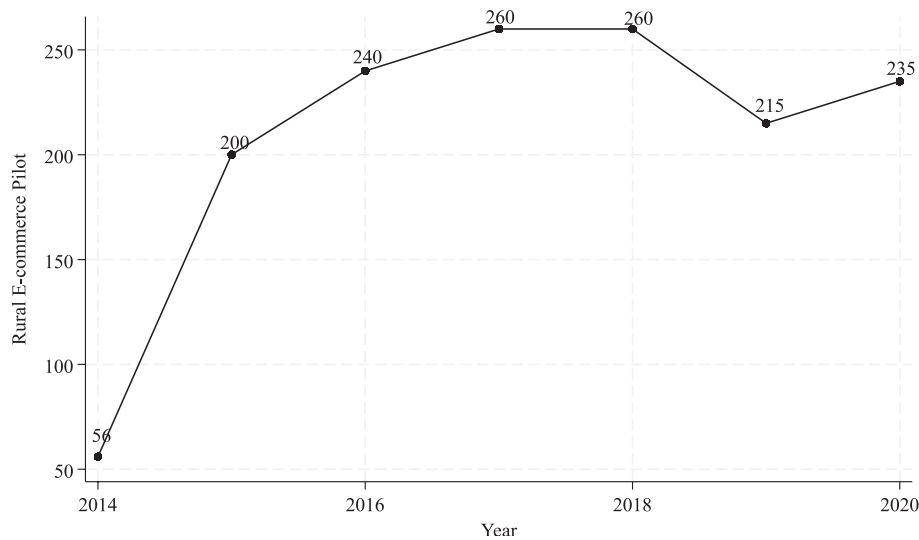


FIGURE 1 | Number of rural e-commerce pilots.

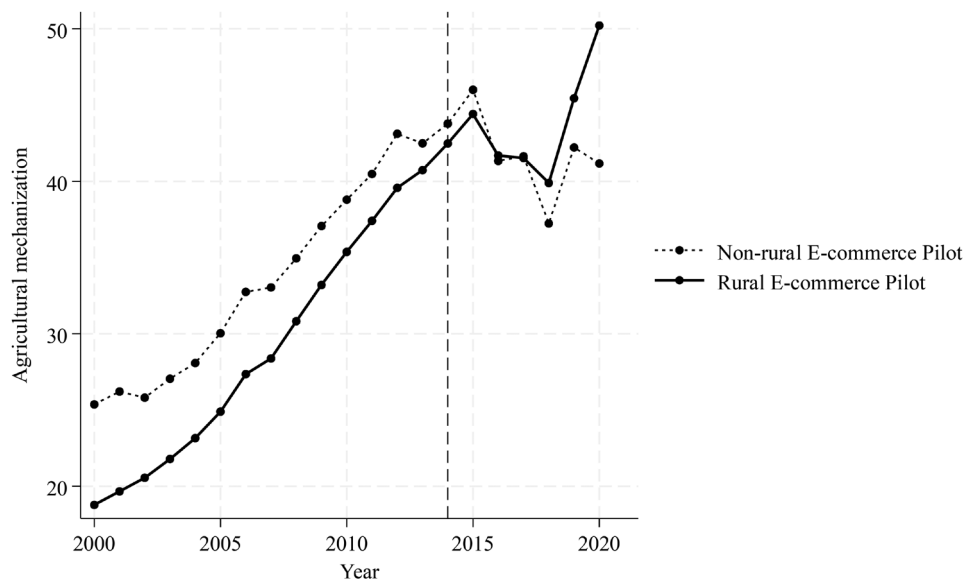


FIGURE 2 | Policy effect diagram.

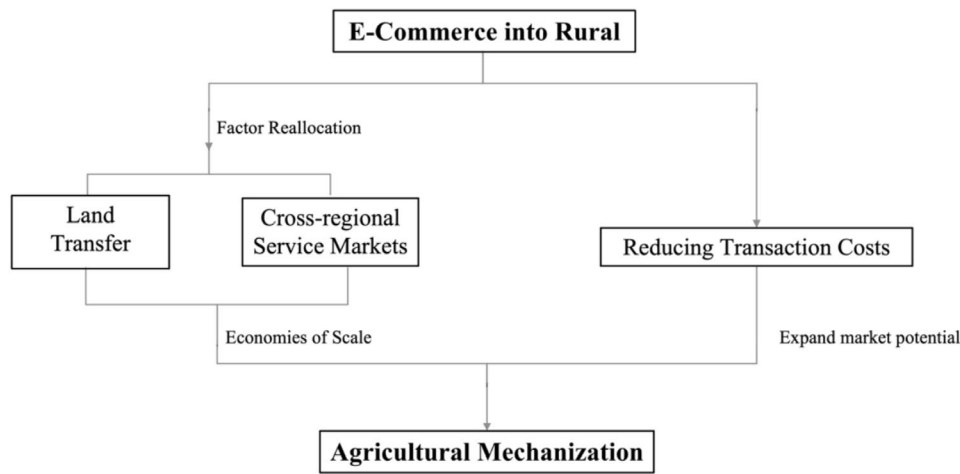


FIGURE 3 | Theoretical framework.

limitations of traditional markets (Shimamoto et al. 2015), drastically lowering search and matching costs. This connects farmers to a broader consumer base (Leong et al. 2016; Zhang et al. 2022) and fundamentally expands their market potential (Mintert et al. 2003; Parker et al. 2016; Song and Morgan 2019).

This integration of fragmented, local markets into a single, national market is the primary “shock” (Couture et al. 2021). As the sales market expands, farmers face greater and more certain market demand. This creates a powerful financial motivation to scale up production. As increased output is difficult to achieve solely through traditional labor-intensive methods, this generates a direct economic incentive to adopt agricultural machinery to enhance productivity and meet the new market opportunities (Lu et al. 2022). Therefore, we obtain the first hypothesis:

H1. *Rural e-commerce promotes the agricultural mechanization level by reducing transaction costs.*

But the new incentive to “scale up” immediately confronts a primary barrier: the nature of agricultural capital. Mechanization is characterized by capital indivisibility and high fixed costs (Zhang et al. 2017).

For a smallholder farmer operating on fragmented plots—a defining feature of Chinese agriculture (Wang et al. 2020)—the average cost of owning and operating such machinery is prohibitively high. The high fixed cost cannot be amortized over their limited output, making labor-intensive methods the only rational choice, even if they face high market demand. Therefore, for the “demand shock” from e-commerce to translate into mechanization, farmers must find a way to achieve economies of scale. This implies a fundamental reallocation of production factors to solve the scale dilemma. We argue this reallocation occurs through two primary channels.

The first channel involves increasing the farmer’s internal operational scale through land transfers. The heightened profitability driven by e-commerce raises the marginal value product of land, stimulating demand for land rental among high-ability farmers and new agricultural entities. E-commerce can further promote land transfer by enhancing information and trust within the village. By renting in land, these motivated producers consolidate

scattered plots, expanding their operational area from subsistence size to a scale where machine ownership becomes financially rational (Wang et al. 2020). The increase in scale makes the investment profitable, thus promoting mechanization adoption.

H2. *Rural e-commerce promotes agricultural mechanization by facilitating land transfers.*

The second channel provides a solution for smallholders who are unable or unwilling to expand their land holdings. This mechanism, based on access over ownership (Zhang et al. 2017), relies on the emergence of a specialized mechanization service market.

In this structure, a machine owner purchases the tractor (bearing the fixed cost) and achieves external economies of scale by servicing hundreds of small farms. E-commerce and related mobile platforms act as catalysts by significantly reducing the search, matching, and coordination costs between service providers and smallholder demand (Yang et al. 2013). This development of cross-regional mechanization services allows small farmers to efficiently purchase the capital service at a variable cost, without having to bear the machine’s high fixed cost. This market solution effectively allows the entire agricultural system to benefit from mechanization.

H3. *Rural e-commerce promotes agricultural mechanization by fostering the development of cross-regional mechanization service markets.*

Together, these hypotheses outline a sequential process: e-commerce expands market access and raises the returns to scale; farmers then overcome capital indivisibility either through land consolidation (internal scale) or through service markets (external scale), ultimately driving mechanization.

4 | Model and Data

4.1 | Empirical Model

To test the impact of the “E-Commerce into the Rural Comprehensive Demonstration County Pilot” (hereinafter referred to as “E-Commerce into Rural”) on agricultural

TABLE 1 | Summary statistics.

	(1)	(2)	(3)	(4)	(5)
Variables	N	Mean	Std dev	Min	Max
Agricultural mechanization	17,803	43.3	40.86	0	336
Rural e-commerce pilot	17,803	0.15	0.35	0	1
Economic development	17,803	13.78	1.114	9.327	17.57
Human capital	17,803	10.6	0.928	6.492	13.05
Financial development	17,803	13.36	1.182	5.288	17.02
Informatization level	17,803	10.39	1.162	2.565	15.47
Industrial structure	17,803	1.42	1.079	0.0182	17.28

mechanization, we use the staggered DID model with year and county fixed effects:

$$y_{it} = \beta_1 did_{it} + C\beta_C + \sigma_i + \mu_t + \varepsilon_{it} \quad (1)$$

The dependent variable y_{it} is total agricultural machinery power for i county in year t . The core independent variable did is designated as a rural e-commerce pilot for “E-Commerce into Rural”. If it is selected as a rural e-commerce pilot in a certain year or later, it is coded as 1; otherwise, it is coded as 0. C is a set of county-level control variables, including economic development, human capital, financial development, informatization level, and industrial structure. Economic development is measured by the natural logarithm of regional gross domestic product (GDP). Human capital is quantified by the natural logarithm of the number of students in primary and secondary schools. Financial development is gauged by the natural logarithm of savings deposits. Informatization level is assessed by the natural logarithm of the number of telephone users. Industrial structure is the ratio of value added in the secondary industry to that in the tertiary industry. σ_i are county fixed effects, and μ_t are year fixed effects. Standard errors clustered at the county-level are in parentheses. The detailed variables definitions are shown in Table A1.

4.2 | Data

The data utilized in this study consists of two parts: macro data and micro data. This approach is adopted due to the limitations of micro-level data in accurately measuring the level of agricultural mechanization, and the incompleteness of county-level data, which hinders a clear understanding of the impact mechanism of the “E-Commerce into the Rural Comprehensive Demonstration County Plan” policy on agricultural mechanization. Therefore, we combine the strengths to conduct comprehensive research.

The macro data comprise panel data at the county level, sourced from the “China county statistical yearbook”. To exclude the influence of the agricultural machinery purchase subsidy policy, which achieved full coverage at the county level in 2009. So, the sample used in this study consists of non-balanced panel data from 2010 to 2020, covering 2076 counties and totaling 17,803 county-level samples. Table 1 presents the macro data used in

this study. Among them, the level of agricultural mechanization that we focus on is measured by the total power of agricultural machinery. The unit is 10,000kW. This index is summarized by the total power of machinery and equipment actually used by each household in agricultural production. Compared with other indicators, such as the ownership of agricultural machinery, this indicator includes cross-district operations and can better reflect the actual situation of the level of agricultural mechanization.

Micro data is derived from the CFPS, a database that tracks and collects data at the individual, household, and community levels, reflecting changes in Chinese society, economy, population, education, and health (Xie and Hu 2014). This survey includes inquiries about household agricultural machinery usage and leasing, providing a micro data foundation for studying the impact mechanism.⁴ Data from 2012, 2014, 2016, 2018, and 2020 are primarily integrated for mechanism analysis. Data from 2010 are not utilized mainly due to the absence of relevant agricultural machinery-related questionnaires that year.

5 | Empirical Result

5.1 | Baseline

Table 2 presents the baseline estimation results based on Equation (1). To ensure robustness, we progressively introduce control variables. As shown in columns (1) and (2), the coefficients of interest are significant positive at the 1% level. According to the preferred specification in Column (2), the implementation of the “E-Commerce into Rural” policy led to an average increase of 26,700kW in total agricultural machinery power for demonstration counties, which translates to a sizable effect of 6.16% relative to the sample mean.

This empirical finding can be interpreted through a clear demand-side mechanism established in the literature. The core logic is that rural e-commerce expands market potential by breaking geographical constraints of traditional markets, thereby connecting farmers to a broader consumer base (Leong et al. 2016; Zhang et al. 2022). Facing this enlarged and more certain market demand (Mintert et al. 2003; Parker et al. 2016; Song and Morgan 2019), agricultural producers are financially

TABLE 2 | The effects of the “E-Commerce into Rural” on agricultural mechanization.

Variables	(1)	(2)
	Agricultural mechanization	Agricultural mechanization
Rural e-commerce pilot	2.997*** (0.605)	2.666*** (0.579)
County-level controls	No	Yes
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	17,803	17,803
Adjusted R-squared	0.940	0.940

Note: This table reports the effects of the “E-Commerce into Rural” on agricultural mechanization. Column (1) doesn’t have control variables. Column (2) has control variables. All specifications control for year fixed effects and county fixed effects. Standard errors clustered at the county-level are in parentheses. ***, **, and * indicate p value below 0.01, 0.05, and 0.1, respectively.

motivated to scale up production. As increased output is difficult to achieve solely through labor-intensive methods, this creates a direct economic incentive to adopt agricultural machinery to enhance productivity and meet the new market opportunities (Lu et al. 2022). The significant increase in machinery power we detect is the direct manifestation of this investment response. This result provides initial, aggregate evidence that the “E-Commerce into Rural” policy indeed stimulates agricultural mechanization by altering fundamental market incentives.

As shown in Table A2, we examine the impact on different types of agricultural machinery using provincial-level data. We find that the expansion of e-commerce into rural areas significantly increases the adoption of small tractors and their complementary equipment, but has no significant effect on large tractors and their complements. This indicates that the e-commerce initiative primarily benefits small-scale farming rather than large agricultural collectives. These findings suggest that this policy may offer valuable insights for developing countries where smallholder farming is prevalent.

5.2 | Robustness Test

In the robustness test, we mainly test the model settings and solve the data problems. We use the staggered DID model, which requires the same ex-ante trend and no heterogeneous treatment effects. In addition, we check whether the selection of control group and experimental group is random. Therefore, in this paper, methods such as parallel trend test, placebo test, heterogeneous-robust DID estimation, Bacon decomposition, CD decomposition, PSM and control time trend are used to test the model settings. The data used in this paper are unbalanced panel data. That’s because the key explained variables are missing. Finally, to verify the robustness of the results, controlling other policy effects, using one-period lagged control variables, and using intensity DID models are eliminated.

5.2.1 | Parallel Trend Test

To ensure the validity of staggered DID estimation, we employ the event study approach to test the parallel trend assumption

and examine the dynamic evolution of the policy’s impact. The empirical specification is as follows:

$$y_{it} = \sum_{k \neq -1} \beta_k D_{it}^k + C\beta_C + \sigma_i + \mu_t + \varepsilon_{it} \quad (2)$$

where D_{it}^k are a series of dummy variables representing the k th year relative to the initial implementation of the “E-Commerce into Rural” policy. Following Friedt and Toner-Rodgers (2022), we set the event window to 6 years before and 6 years after the treatment, with the year prior to the intervention ($k = -1$) serving as the reference period to avoid multicollinearity. Notably, we exclude observations from 2013 in the pre-treatment period to mitigate potential confounding effects from the major adjustment of agricultural machinery subsidy policies in that year.

The results of the event study are plotted in Figure 4. We observe two critical patterns that validate the identification strategy. Firstly, all coefficients for the periods prior to the policy ($k < -1$) are statistically insignificant and close to zero, with no discernable trend. This indicates that before the “E-Commerce into Rural” initiative, there were no systematic differences in agricultural mechanization trends between the treatment and control counties, effectively ruling out concerns regarding self-selection bias or pre-existing trends.

Secondly, following the policy intervention ($k \geq 0$), the coefficients become significantly positive and exhibit an evident upward trajectory over time. This suggests that the policy does not yield a mere one-time shift but rather a cumulative and sustained “pull effect” on agricultural mechanization. The widening gap in the later periods reflects the time-lagged nature of infrastructure construction and the gradual maturation of rural e-commerce ecosystems, which progressively lowers the barriers to technology adoption for smallholders.

5.2.2 | Placebo Test

To further ensure that the baseline results are not driven by unobserved time-varying factors or idiosyncratic county-level shocks, we perform a rigorous placebo test. We implement a randomized treatment assignment procedure by simulating

a “virtual” policy intervention. Specifically, we randomly select a subset of counties from the sample to serve as a pseudo-treatment group and assign them with pseudo-implementation years. This permutation process is repeated 1000 times to generate a kernel density distribution of estimated coefficients under the null hypothesis—namely, that the “E-Commerce into Rural” policy has no systematic effect on agricultural mechanization.

The results of the simulation, as illustrated in Figure 5, reveal that the distribution of these placebo coefficients is centered near zero and closely follows a normal distribution. Furthermore, the p values for the vast majority of these 1000 iterations are well above the 0.1 significance level, indicating that the randomly assigned policies possess no explanatory power. In stark contrast, the actual baseline estimate (indicated by the vertical dashed line) sits far in the right tail of the distribution as a clear outlier. This statistical divergence

demonstrates that the probability of the primary results being a product of chance is extremely low. Consequently, this test effectively rules out the possibility that the identified positive impact is driven by omitted variables, thereby confirming the robustness of causal inference.

5.2.3 | Heterogeneity-Robust Estimator

Due to the non-random selection of demonstration counties in terms of their geographical locations and potential variations in the timing of specific implementation actions, the treatment effects of the “E-Commerce into Rural” initiative may exhibit regional or temporal heterogeneity. It leads to biased coefficients in the two-way fixed-effects regression. Following Abadie (2005), Sant’Anna and Zhao (2020) and de Chaisemartin and D’Haultfœuille (2023), we employ

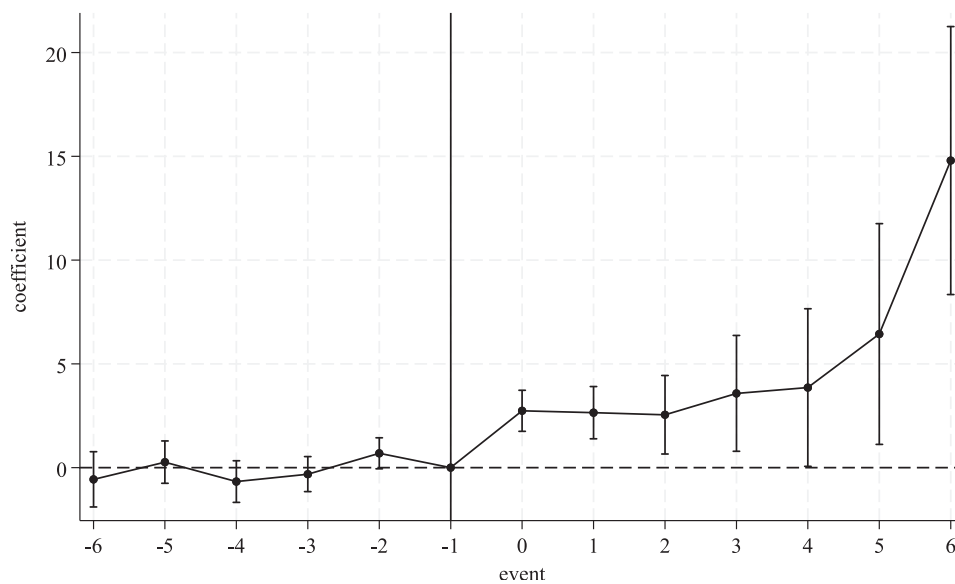


FIGURE 4 | Parallel trend test. Note: Dots are coefficients and segment lengths are 95% confidence intervals.

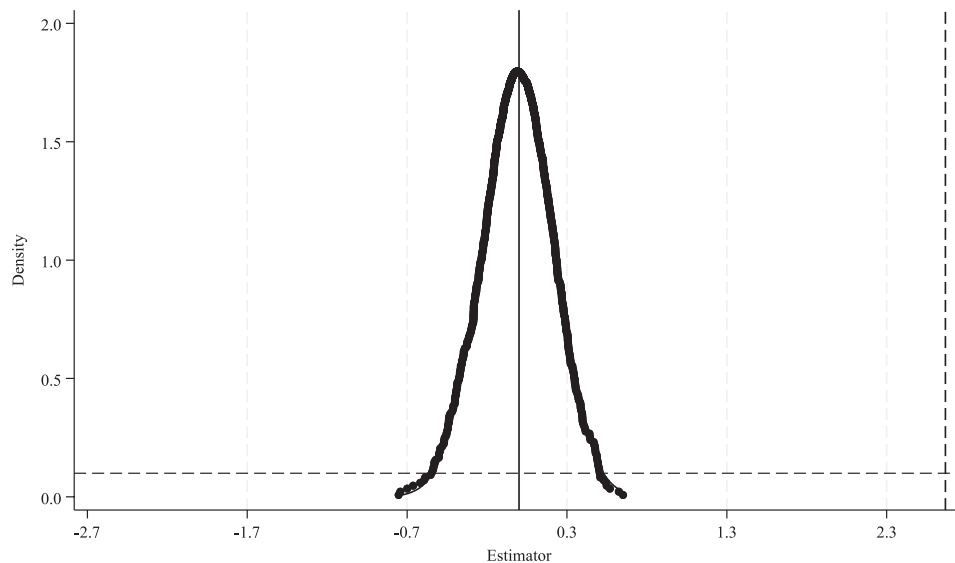


FIGURE 5 | Placebo test. Note: The solid line is the result of one thousand simulations, and the dashed line is the result of the baseline regression.

heterogeneity-robust DID estimation methods. The regression results present in Panel A in Table 3. Columns (1–4), respectively, reflect the doubly robust DID estimator based on stabilized inverse probability weighting and ordinary least squares (Dripw), the doubly robust DID estimator based on inverse probability of tilting and weighted least squares (Drimp), the inverse probability weighting DID estimator with stabilized weights (Stdipw), and the inverse probability weighting DID estimator (Ipw). At a significance level of 1%, the average treatment effects are consistently positive, indicating that even in the presence of heterogeneous treatment effects, the “E-Commerce into Rural” initiative significantly promotes the level of agricultural mechanization.

5.2.4 | Negative Weighting Problems

Given the staggered nature of the “E-Commerce into Rural” policy, the traditional two-way fixed effects (TWFE) estimator may be biased if treatment effects vary over time. To address this concern, we perform a Goodman-Bacon decomposition to examine the composition of the DID estimator. This method identifies the TWFE estimator as a weighted average of three distinct components: (1) treated groups compared to never-treated groups, (2) early-treated groups compared to later-treated groups (where the latter serve as controls), and (3) later-treated groups compared to early-treated groups (where the former serve as controls). The latter comparison is often considered problematic, as the “already-treated” group may not provide an appropriate counterfactual, potentially leading to inaccurate or even negative weights.

To satisfy the requirements for Bacon decomposition, we utilize a balanced panel of 2076 counties from 2010 to 2017. The results of the decomposition are visualized in Figure 6 and summarized as follows: Most of the weight in the estimation is attributed to the “Treated vs. Never Treated” comparison. Specifically, this robust comparison yields a coefficient of 2.9795 and accounts for 86% of the total weight. In contrast, the potentially biased comparison (later vs. already treated) yields an estimated effect of -57.47 , but its relative weight is negligible at only 0.5%.

The diagnostic results in Figure 6 provide strong evidence for the reliability of baseline findings. Although the coefficient for the inappropriate comparison is numerically large, its minimal weight ensures that it does not significantly distort the overall estimation. These findings effectively rule out severe negative weighting issues, confirming that the identified positive impact of e-commerce on mechanization is not an artifact of heterogeneous treatment effects across different cohorts.

5.2.5 | Non-Random Selection of Demonstration Counties

Due to the potential sample selection bias in the selection of demonstration counties, where impoverished counties or revolutionary base areas may be more likely to become demonstration

counties, we employ the PSM-DID method and add time pre-trend terms to solve this issue.

The PSM-DID result shows in the Column (1) in Panel B in Table 3. By matching based on economic level, human capital, financial development, and informatization level, and subsequently regressing the matched results, the coefficients remain significantly positive at the 1% significance level. Additionally, we control county-level pre-trend including poor county and revolutionary base area. Coefficients are all significantly positive in the columns (2–4) in Panel B in Table 3. This indicates that even after mitigating the sample selection bias issue, the coefficients remain significantly positive. Although using PSM-DID, we can only solve selection bias caused by observed variables. Because the selection of demonstration counties is focused on poor counties, the results just underestimate the effect of rural e-commerce.

5.2.6 | Unbalanced Panel Data

We use the unbalanced panel data to cause sample incompleteness, so we use balanced panel data to regress and Heckman two-stage way to solve it. Initially, considering substantial data gaps after 2017, the sample is confined to balanced panel observations spanning from 2010 to 2017 for regression analysis. Results, as delineated in Column (1) of Panel C in Table 3, persist in revealing statistically significant positive coefficients at the 1% significant level.

The principal source of data incompleteness in this study stems from the absence of data on agricultural mechanization levels. However, recognizing that missing data may emanate from counties with lower levels of agricultural mechanization, there exists a potential for baseline regression to overstate the impact of rural e-commerce on agricultural mechanization. To mitigate this concern, the Heckman two-stage method is employed. Given the intimate linkage between agricultural mechanization levels and agricultural development, the value added of the primary industry is incorporated as a selection variable. Regression outcomes show in Column (2) of Panel C in Table 3. It exhibits a significantly positive coefficient for the inverse Mills ratio (Inv mills) at the 1% level, suggestive of self-selection bias. Following the adjustment for the inverse Mills ratio, the coefficient is still significantly positive at the 1% significant level. Thus, it proves the robustness of the conclusions.

5.2.7 | Control Other Policies

During the implementation of rural e-commerce initiatives, several other policies related to the digital economy were concurrently enacted. We present a comprehensive examination of Chinese digital economy policies from 2010 to 2020. It includes policies such as the National Big Data Comprehensive Pilot Zone, information consumption trials, National Pilot Policy of Information Benefiting the People, and Broadband China. To mitigate the potential influence of these additional policies, we control for their effects in the regression analysis, as presented in Panel D in Table 3.

TABLE 3 | Robustness test.

Panel A: Heterogeneity-robust estimator				
	(1)	(2)	(3)	(4)
	Dripw	Drimp	Stdipw	Ipw
ATT	2.312*** (0.821)	1.891*** (0.641)	1.906*** (0.630)	1.960*** (0.651)
Panel B: Non-random selection of demonstration counties				
	(1)	(2)	(3)	(4)
	PSM-DID	Time pre-trend terms		
Variables	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization
Rural e-commerce pilot	2.591*** (0.581)	1.765*** (0.555)	1.536*** (0.565)	1.536*** (0.565)
County-level pre-trend $\times T$	No	Yes	Yes	Yes
County-level pre-trend $\times T^2$	No	No	Yes	Yes
County-level pre-trend $\times T^3$	No	No	No	Yes
County-level controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	17,729	17,803	17,803	17,803
Adjusted <i>R</i> -squared	0.940	0.942	0.942	0.942
Panel C: Other robustness test				
	(1)	(2)	(3)	(4)
	Balanced panel data	Heckman	Lagged One-period Control Variable	DID with Intensity Index
Variables	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization
Rural E-commerce pilot	2.541*** (0.554)	2.490*** (0.569)	2.480*** (0.583)	2.274*** (0.447)
Invmills		2.253*** (0.243)		
County-level controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	13,360	17,803	17,384	17,803
Adjusted <i>R</i> -squared	0.957	0.942	0.941	0.940
Panel D: Control other policies				
	(1)	(2)	(3)	(4)
Variables	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization
Rural e-commerce pilot	2.556*** (0.570)	2.522*** (0.565)	2.523*** (0.565)	2.529*** (0.564)

(Continues)

TABLE 3 | (Continued)

Variables	Panel D: Control other policies			
	(1)	(2)	(3)	(4)
	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization
National big data comprehensive pilot zone	Yes	Yes	Yes	Yes
National pilot policy of information benefiting the people	No	Yes	Yes	Yes
Information consumption trials	No	No	Yes	Yes
Broadband China	No	No	No	Yes
County-level controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	17,803	17,803	17,803	17,803
Adjusted R-squared	0.942	0.942	0.942	0.942

Note: This table reports the results of robustness test of baseline regression. In Panel A, we use a heterogeneity-robust estimator to check whether the treatment effects of the “E-Commerce into Rural” initiative may exhibit regional or temporal heterogeneity (Abadie 2005; Sant’Anna and Zhao 2020; de Chaisemartin and D’Haultfoeuille 2023). Columns (1–4), respectively, reflect the doubly robust DID estimator based on stabilized inverse probability weighting and ordinary least squares (Dripw), the doubly robust DID estimator based on inverse probability of tilting and weighted least squares (Drimp), the inverse probability weighting DID estimator with stabilized weights (Stdipw), and the inverse probability weighting DID estimator (Ipw). In Panel B, we use the PSM-DID method and add time pre-trend terms to solve the potential sample selection bias in the selection of rural E-commerce pilots. The PSM-DID result shows in Column (1). Additionally, we control for county-level pre-trends including poor county and revolutionary base area in Columns (2–4). In Panel C, we do other robustness tests. In Columns (1) and (2), we use balanced panel regression and Heckman two-stage estimation techniques to solve the problems of the unbalanced panel data, such as the absence of data on agricultural mechanization levels. In Column (3), we use a lagged one-period control variable approach to address the issue of potential endogeneity arising from the presence of bad control variables. In Column (4), we use the DID with Intensity Index model to solve the issue of repeated selection of demonstration counties. In Panel D, we control for other policies, such as the National Big Data Comprehensive Pilot Zone, information consumption trials, National Pilot Policy of Information Benefiting the People, and Broadband China. Columns (1–4) are results of progressively incorporating these policies as control variables. All specifications control for year fixed effects and county fixed effects. Standard errors clustered at the county-level are in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

By progressively incorporating these policies as control variables and isolating their impacts, the coefficient associated with rural e-commerce remains significantly positive even after accounting for these policy influences. This affirms the robustness of the findings.

5.2.8 | Lagged One-Period Control Variable

Bad control variables cause the issue of potential endogeneity. For example, when both the explanatory and dependent variables may influence the control variables, it leads to spurious significance (Cinelli et al. 2022). Therefore, we employ a lagged one-period control variable approach to address it. This way aims to mitigate the risk of spurious causal relationships induced by reverse causality. As demonstrated in Column (3) of Panel C in Table 3, even after lagging the control variables by one period, the coefficients remain significantly positive at the 1% significant level, thus further corroborating the robustness of the results.

5.2.9 | DID With Intensity Index

To address the issue of repeated selection of demonstration counties, which suggests a relatively higher level of government support, the DID with intensity index model is employed in this paper. Utilizing the frequency of selection of demonstration

counties as an intensity variable, the regression analysis reveals that the coefficient of the intensity DID is significantly positive at the 1% significance level, thereby confirming the robustness of the results.

5.3 | Mechanism Analysis

The baseline regression results demonstrate that rural e-commerce promotes agricultural mechanization by expanding demand, yet the specific mechanisms remain underexplored. In the mechanism analysis section, we use both macro- and micro-level data to examine whether demand expansion driven by e-commerce leads farmers to increase other factor inputs. First, we verify the importance of reduced transaction costs. Next, we consider inputs related to land, as agricultural mechanization often requires economies of scale achieved through land expansion (Wang et al. 2020). Finally, we examine the role of cross-regional operations, which serve as an important pathway to agricultural mechanization in developing countries, especially those characterized by smallholder economies (Zhang et al. 2017).

5.3.1 | Transaction Costs

The two main sources of transaction costs or market access barriers in rural e-commerce are digital literacy and logistics

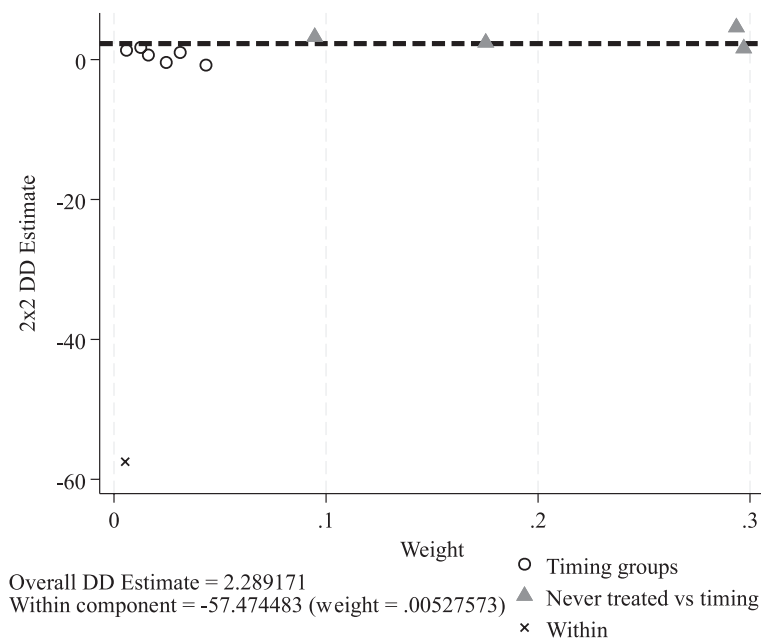


FIGURE 6 | Bacon decomposition.

constraints (Shimamoto et al. 2015). Therefore, we further examine whether digital literacy or logistics limitations in counties attenuate the policy effects by including interaction terms, with results presented in Table 4.

As shown in Column (1), we first interact the level of digital infrastructure with the policy dummy (DID) and find that the interaction term is significantly positive, indicating that digital infrastructure moderates the policy impact. Subsequently, we use topographic relief as a proxy for logistics conditions—greater relief indicates more rugged terrain and higher transport difficulty (Lei and Chen 2025; Chen 2025). A dummy variable is constructed based on the median national topographic relief: counties with relief below the median are classified as having better logistics conditions (assigned 1), otherwise 0. The results in Column (2) show that terrain barriers significantly influence the effect of e-commerce rural promotion, suggesting that logistics infrastructure must be improved alongside e-commerce initiatives.

In summary, we find that e-commerce rural promotion mainly reduces transaction costs through digital technologies, yet its effectiveness remains constrained by logistics conditions.

5.3.2 | Land Transfer

The primary barrier to agricultural mechanization lies in its high operational cost. Agricultural machinery represents a fixed capital investment with substantial upfront expenditures. Achieving economies of scale through land consolidation is a critical pathway to making this investment viable, with land transfer serving as the enabling market institution.

To test this mechanism, we first examine the impact of e-commerce on the activity of the land rental market. Data

TABLE 4 | Reducing transaction costs.

Variables	(1)	(2)
	Agricultural mechanization	Agricultural mechanization
Rural e-commerce pilot	-3.807 (3.779)	3.820*** (1.050)
Rural e-commerce pilot × informatization	0.647* (0.385)	
Rural e-commerce pilot × logistics		-1.937* (1.118)
County-level controls	Yes	Yes
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	17,803	17,803
Adjusted R-squared	0.940	0.940

Note: This table reports the mechanism of reducing transaction costs effect. All specifications control for year fixed effects and county fixed effects. Standard errors clustered at the county-level are in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

on the number of land transfers are sourced from the China Academy for Rural Development-Qiyang China Agri-research Database (CCAD) at Zhejiang University. These records, aggregated to the county level, measure annual land transfer transactions. As shown in Column (1) of Table 5, the “E-Commerce into Rural” pilot policy increases the number of land transfers at the 5% significance level. This indicates that rural e-commerce reduces transaction costs and stimulates

TABLE 5 | The effects of the “E-Commerce into Rural” on land transfer.

Variables	(1)	(2)	(3)	(4)	(5)
	Macro data			Micro data	
	Land transfers	Sown area	Area harvested by machine	Whether land is rented	
Rural e-commerce pilot	0.462** (0.220)	1.209** (0.558)	0.105*** (0.023)	0.033*** (0.007)	−0.049*** (0.012)
Rural e-commerce pilot × rural					0.099*** (0.012)
County-level controls	Yes	Yes	Yes	Yes	Yes
County/Family fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	17,803	9959	9552	31,649	29,901
Adjusted R-squared	0.189	0.989	0.938	0.359	0.355

Note: This table reports the effects of the “E-Commerce into Rural” on land transfer. Columns (1–3) use macro data and control year fixed effects and county fixed effects, and Columns (3) and (4) use micro data and control year fixed effects and family fixed effects. In Column (1), the dependent variable is *Land transfers*, which is equal to the number of land transfers at the county level. In Column (2), the dependent variable is *Sown area*, which is equal to the log of the total sown area of crops at the county level. In Column (3), the dependent variable is *Area harvested by machine* at the county level. In Columns (3) and (4), the dependent variable is *whether land is rented* at the family level. Column (5) is based on Column (4) adding cross-multiplier terms for rural and whether it is the rural E-commerce pilot. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

the land rental market, creating conditions for potential scale expansion (Zhang et al. 2018).

We further test whether land transfer is a key channel through which the policy promotes mechanization. In Table A3, we introduce an interaction term between the policy dummy and the land transfer measure. The coefficient is positive and significant at the 1% level, suggesting that the mechanization-enhancing effect of e-commerce is stronger in counties with more active land markets. This supports the hypothesis that the policy operates partly through facilitating land transfer.

An active rental market, however, does not automatically imply an increase in operational farm size. To verify that the stimulated transactions lead to actual scale expansion, we test the policy's effect on the total sown area. Using the logarithm of the total crop sown area from county yearbooks as the dependent variable, Column (2) of Table 5 shows that the policy raises the sown area at the 5% significance level. This confirms that the activation of the land market ultimately results in larger production units.

Finally, we examine whether this scale expansion directly increases mechanization. Using the logarithm of the machine-harvested area as the outcome, Column (3) of Table 5 demonstrates that the pilot policy significantly increases mechanized area at the 5% level. This completes the empirical chain: the policy reduces transaction costs, activates the land rental market, leads to the consolidation of operational scale, and thereby promotes the adoption of agricultural machinery. We supplement this macro-level analysis with household-level evidence from the CFPS. Using a dummy variable for whether a household rents in land as the dependent variable, we control for household and county characteristics and include

household and year fixed effects. Household-level controls include household size and the logarithm of net income. Column (4) of Table 5 shows that rural e-commerce increases the probability of land rental at the 1% significance level. In Column (5), we add an interaction between a rural household dummy and the policy dummy. The positive and significant coefficient (1% level) indicates that the policy has a particularly strong effect on incentivizing land rental among rural households, further reinforcing the land-transfer channel. In summary, facing an expanded product demand shock facilitated by e-commerce, farmers are motivated to enlarge their operational scale through the land rental market. This consolidation allows them to overcome the capital indivisibility of agricultural machinery by achieving economies of scale, which in turn drives the adoption of mechanization.

5.3.3 | Cross-Regional Mechanization Harvest

For smallholder economies, the high fixed cost of purchasing agricultural machinery is often prohibitive. Leasing machinery services presents a viable alternative, allowing farmers to access mechanization without the capital burden of ownership. By facilitating a market for cross-regional machinery operations, this “access-over-ownership” model can significantly lower the effective cost of mechanization (Zhang et al. 2017).

We first examine this channel at the household level using data from the CFPS. We construct a binary variable indicating whether a household incurred any rental cost for agricultural machinery. Since cross-regional operations are essentially the leasing of agricultural machinery, we use this as a proxy variable for cross-regional operations at the micro level. In the context of smallholder farming, formal rentals are frequently the means through which

TABLE 6 | The effects of the “E-Commerce into Rural” on cross-regional mechanization harvest.

Variables	(1)
	Micro data
	Rural samples
	Rental of agricultural machinery
Rural e-commerce pilot	0.011*** (0.003)
County-level controls	Yes
Family fixed effects	Yes
Year fixed effects	Yes
Observations	31,649
Adjusted R-squared	0.119

Note: This table reports the effects of the “E-Commerce into Rural” on cross-regional mechanization harvest. We use micro data and control year fixed effects and family fixed effects. The samples only include rural families. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

farmers access services provided by larger, mobile operators who work across regions. The regression results, presented in Table 6, show that the “E-commerce into Rural” policy significantly increases the probability of machinery rental among rural households. This suggests the policy stimulates demand for mechanization services, reducing the private cost of adoption.

To directly test for cross-regional spillovers and market integration, we employ a Spatial Durbin Model (SDM) following the framework of Wang et al. (2023). This approach explicitly accounts for spatial interdependence, allowing us to decompose the policy's impact into a direct effect on the local county and an indirect (spillover) effect on neighboring counties. The specification is as follows:

$$y_{it} = \beta_1 did + \beta_2 C + \rho WY_{t-1} + \theta WX_{t-1} + \sigma_i + \mu_t + \varepsilon_{it} \quad (3)$$

where W denotes the spatial weight matrix. The parameter ρ captures the spatial lag of the dependent variable, representing the spillover effect of agricultural mechanization from neighboring regions. The coefficient θ reflects the impact of the “E-Commerce into Rural” policy in neighboring counties on local mechanization. In this setting, the total impact of the policy can be decomposed into a direct effect (the impact on the local county) and an indirect effect (the spatial spillover to adjacent counties). A significantly positive θ and ρ would provide robust evidence that rural e-commerce promotes cross-regional mechanization harvests, thereby enhancing the overall mechanization level through spatial integration.

We estimate the model using a balanced county-level panel (2010–2017). A second-order inverse distance matrix is used as the primary spatial weight, with a first-order matrix for robustness checks. Diagnostic tests confirm positive spatial autocorrelation (Moran's I is significant at the 1% level for mechanization) in Table A4, justifying the spatial econometric approach.

The SDM results are shown in Table 7. Consistent with the baseline findings, the direct effect of the policy remains positive and significant. Crucially, we find a statistically significant positive indirect effect (spatial spillover), with both ρ and θ positive and significant. This indicates that the policy fosters a synchronized regional market for machinery services: e-commerce adoption in one county not only increases local mechanization but also generates positive spillovers by enhancing service availability in adjacent counties.

In summary, the evidence supports a cross-regional service channel. By reducing search and coordination costs, rural e-commerce helps cultivate a shared market for agricultural machinery services. This allows smallholders to overcome capital constraints by purchasing mechanization as a variable-cost service rather than a fixed asset, thereby promoting adoption through spatial integration and external economies of scale.

5.4 | Heterogeneity Analysis

To further explore the distributional impacts of the “E-Commerce into Rural” policy, we analyze whether its effects vary across regions with different land endowments and agricultural productivity. This allows us to assess whether e-commerce acts as a “leveling” force that benefits disadvantaged agricultural areas or if it disproportionately favors well-endowed regions.

Columns (1) and (2) of Table 8 report the heterogeneous effects based on administrative area size. We categorize counties into two groups using the median administrative area as the threshold. The results reveal that the policy's impact is predominantly driven by counties with smaller administrative areas, where the coefficient is significantly positive at the 1% level. In contrast, the effect in counties with larger administrative areas is statistically insignificant.

From a distributional perspective, this finding is particularly noteworthy. Smaller administrative areas are typically characterized by greater land fragmentation and a higher density of smallholders, who historically faced the highest barriers to mechanization due to capital indivisibility. Our results suggest that e-commerce provides a vital mechanism for these fragmented regions to overcome the “small-scale trap.” By facilitating land transfer and consolidating plots, e-commerce enables smallholders in these constrained areas to reach the minimum efficient scale required for machinery adoption (Wang et al. 2020), whereas larger regions may have already achieved this scale through traditional means.

We further investigate the policy's impact across different levels of agricultural development. Following the classification of China's grain-surplus provinces, we divide the sample into high grain output (surplus) and low grain output (deficit) regions. As shown in columns (3) and (4) of Table 8, the coefficients for high-output regions—primarily Inner Mongolia, Heilongjiang, Jilin, Henan, and Anhui—are insignificant. However, the impact on low-output regions is positive and significant at the 1% level.

This divergence highlights the inclusive nature of digital dividends. High grain-output regions usually possess established agricultural infrastructure and higher baseline mechanization

TABLE 7 | The spatial model.

Variables	(1)	(2)
	Second-order inverse distance matrix	First-order inverse distance matrix
	Agricultural mechanization	Agricultural mechanization
Rural e-commerce pilot	0.764*** (0.258)	1.734*** (0.285)
θ	1.815** (0.822)	1.519** (0.605)
ρ	0.990*** (0.003)	0.896*** (0.010)
County-level controls	Yes	Yes
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	13,360	13,360
R-squared	0.069	0.001

Note: This table reports the spillover effects of the “E-Commerce into Rural” on agricultural mechanization through the spatial model. Columns (1–3) are the OLS model, the spatial Durbin model using the second-order inverse distance matrix and the first-order inverse distance matrix, respectively. Standard errors clustered at the county-level are in parentheses. ***, **, and * indicate p value below 0.01, 0.05, and 0.1, respectively.

TABLE 8 | Heterogeneity analysis.

Variables	(1)	(2)	(3)	(4)
	Small administrative area	Large administrative area	High grain output regions	Low grain output regions
	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization	Agricultural mechanization
Rural e-commerce pilot	2.468*** (0.897)	−0.394 (0.631)	0.747 (1.368)	2.692*** (0.567)
County-level controls	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	8860	8861	3382	14,345
Adjusted R-squared	0.942	0.950	0.951	0.935

Note: This table reports the heterogeneity analysis of the “E-Commerce into Rural” on agricultural mechanization. Columns (1) and (2) illustrate the heterogeneity in administrative area. Counties are classified according to the median value of their area. Counties greater than the median value are defined as having a large administrative area, while counties below the median value are defined as having a small administrative area. Columns (3) and (4) represent the heterogeneity of whether a province is a net grain transfer province. Since 2010, the net grain redeployment provinces of China have concentrated in five provinces: Inner Mongolia, Heilongjiang, Jilin, Henan and Anhui. This paper therefore conducts group regressions based on whether counties are in these five net grain transfer provinces. Standard errors clustered at the county-level are in parentheses. ***, **, and * indicate p value below 0.01, 0.05, and 0.1, respectively.

levels, leading to potential diminishing marginal returns from digital interventions. Conversely, low-output regions, which often lag in agricultural modernization, benefit disproportionately from the e-commerce shock. The policy enables these marginalized regions to bypass traditional developmental stages by integrating into cross-regional mechanization service markets (Zhang et al. 2017). Through e-commerce platforms, farmers in low-productivity areas can more easily rent machinery from surplus regions, effectively lowering the entry barrier to mechanization. These findings suggest that rural e-commerce serves

as an equalizer, narrowing the mechanization gap between leading and lagging agricultural regions.

6 | External Validity

Although empirical analysis is grounded in the context of China’s “E-Commerce into Rural” policy, the implications of findings extend far beyond China’s borders. The economic mechanisms we identified—where digital platforms reduce

transaction costs and induce factor reallocation—possess strong external validity for many developing economies facing similar structural constraints.

First, the policy experiment in China is not unique but rather a precursor to a global trend in the digitization of rural areas. The “E-Commerce into Rural” model, characterized by government-led infrastructure investment coupled with private platform participation, is currently being replicated or adapted in various forms across the Global South. Developing nations with large rural populations, such as Egypt, India, and Vietnam, are actively promoting rural e-commerce to integrate smallholders into the modern economy (Couture et al. 2021; Liu et al. 2023). Consequently, the empirical evidence from China serves as a critical reference point for these countries, suggesting that digital interventions can generate positive spillovers into agricultural productivity beyond mere consumption upgrades.

Second, the fundamental structural barrier addressed in this study—the “Smallholder Dilemma”—is a universal feature of agriculture in developing countries. Similar to China, the agricultural sectors in many African and Southeast Asian nations are characterized by land fragmentation, labor-intensive practices, and low productivity (Byres and Byres 1981; Jensen 2007; Benin 2015; Shimamoto et al. 2015; Mohammed et al. 2023). In these contexts, the high fixed costs of machinery (capital indivisibility) typically prevent small farmers from mechanizing. The findings extend this literature by demonstrating that e-commerce does more than just provide price information (Deichmann et al. 2016; Kaila and Tarp 2019); it actively expands market boundaries (Parker et al. 2016). This suggests that the “demand-side shock” we observe in China is a replicable pathway for other agrarian economies to overcome the “smallholder dilemma.”

Finally, the specific pathways we identified—land consolidation and cross-regional mechanization services—offer a roadmap for other nations. Our study shows that full land ownership consolidation is not a prerequisite for mechanization. Instead, e-commerce facilitates a “service-based mechanization” (cross-regional harvest services). This is particularly relevant for countries with rigid land tenure systems. It implies that digital platforms can help other developing countries leapfrog the traditional stage of heavy capital accumulation by promoting a “sharing economy” for agricultural machinery, allowing smallholders to access mechanization services without bearing the prohibitive costs of ownership.

7 | Conclusions

This study provides comprehensive empirical evidence on how digital market integration—specifically through the “E-Commerce into Rural” policy—reshapes agricultural production in a developing country context. Using a staggered DID approach and a decade of county-level data from China, we demonstrate that rural e-commerce has significantly catalyzed agricultural mechanization.

The core contribution of this paper lies in identifying the “market-induced mechanization” mechanism. We find that

e-commerce fundamentally alters the production frontier by reducing transaction costs. This allows for the expansion of land transfer to achieve internal economies of scale and the facilitation of cross-regional mechanization services to achieve external economies of scale. Crucially, the policy exhibits a “pro-poor” or inclusive growth characteristic: its impact is most pronounced in counties with smaller administrative areas and lower baseline grain output—regions that traditionally struggled most with the “Smallholder Dilemma.”

Based on our findings, we propose three targeted policy recommendations for developing countries aiming to harness digital tools for agricultural development:

First, integration of Digital and Logistics Infrastructure: The analysis shows that the benefits of e-commerce are moderated by logistics conditions. Policymakers should not view digital infrastructure in isolation. Investments in broadband must be synchronized with upgrades in rural road networks to ensure that the “virtual” market expansion translates into the physical movement of agricultural goods and machinery.

Second, support for “Service-Oriented” Mechanization: Recognizing that smallholders often lack capital, governments should encourage the development of digital platforms that match machinery owners with small farmers (similar to the “Uberization” of farm equipment). This facilitates the cross-regional service market verified in our study, allowing farmers to access technology without asset ownership.

Third, facilitating Land Rental Markets: Since e-commerce drives the demand for scale, policies should aim to reduce friction in land rental markets. Digital platforms can serve as information hubs for land listings, helping motivated farmers consolidate fragmented plots to achieve the internal economies of scale necessary for mechanization.

Despite the robust findings, this study has limitations that point towards avenues for future research. Primarily, analysis of micro-mechanisms is constrained by data availability. While we utilize county-level macro data and household-level CFPS data, we lack high-frequency, detailed production data at the farm level. Specifically, we cannot fully observe the internal cost structures of households or the specific decision-making process regarding the types of machinery purchased versus rented in response to online orders. Future research relying on more granular survey data could unpack the “black box” of how digital orders specifically alter a farmer’s daily capital-labor substitution decisions. Additionally, examining the impact of e-commerce on agricultural productivity would be a valuable extension of this work.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

- ¹ According to the policy document issued by the Ministry of Commerce of China, we collect it independently.
- ² The data are from the “2023 China Online Retail Market Development Report” released by the Ministry of Commerce of China.
- ³ The fluctuation in total machinery power stems from industrial “de-capacity” and the replacement cycle of obsolete equipment. While the macro-trend reflects the scrapping of outdated units, the divergent growth paths between the two groups underscore the robust impact of the e-commerce policy.
- ⁴ The analysis of the data of restricted districts and counties in this article was carried out in the computer room of the China Social Sciences Survey Center of Peking University.

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Appendix A

TABLE A1 | Variable definitions.

Variable	Definition
Agricultural mechanization	Total agricultural machinery power (10,000 kW).
Rural e-commerce pilot	If it is selected as a rural e-commerce pilot in a certain year or later, it is coded as 1; otherwise, it is coded as 0.
Economic development	The natural logarithm of regional gross domestic product (GDP).
Human capital	The natural logarithm of the number of students in primary and secondary schools.
Financial development	The natural logarithm of savings deposits.
Informatization level	The natural logarithm of the number of telephone users.
Industrial structure	The ratio of value added in the secondary industry to that in the tertiary industry.

TABLE A2 | Disaggregating machinery types.

Variables	(1)		(2)		(3)		(4)	
	Large scale tractors		Small scale tractors		Large scale attached equipment		Small scale attached equipment	
Rural e-commerce pilot numbers	−0.005 (0.004)	Yes	0.008** (0.004)	Yes	0.008 (0.007)	Yes	0.009*** (0.003)	Yes
Province-level controls	Yes		Yes		Yes		Yes	
Province fixed effects	Yes		Yes		Yes		Yes	
Year fixed effects	Yes		Yes		Yes		Yes	
Observations	341		341		341		248	
Adjusted R-squared	0.982		0.975		0.936		0.993	

Note: This table reports the effects of the "E-Commerce into Rural" on different agricultural machinery types. The dependent variables are large scale tractors, small scale tractors, large scale attached equipment, and small scale attached equipment in columns (1–4), respectively. The dependent variables are measured by total machinery power, and the core independent variable Rural E-commerce Pilot number is the number of rural e-commerce pilots in the province. We control for province-level economic development and industrial structure. All specifications control for year fixed effects and province fixed effects. Robust standard errors are in parentheses. ***, **, and * indicate p value below 0.01, 0.05, and 0.1, respectively.

TABLE A3 | Land transfer.

Variables	(1)
	Agricultural mechanization
Rural e-commerce pilot	2.451*** (0.584)
Rural e-commerce pilot × land transfer	0.521*** (0.129)
Province-level controls	Yes
Province fixed effects	Yes
Year fixed effects	Yes
Observations	17,727
Adjusted R-squared	0.940

Note: This table reports the effects of the “E-Commerce into Rural” on agricultural mechanization. All specifications control for year fixed effects and county fixed effects. Standard errors clustered at the county-level are in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

TABLE A4 | Moran's *I*.

Agricultural mechanization				Rural e-commerce pilot			
Year	Moran's <i>I</i>	Sd(<i>I</i>)	<i>p</i>	Year	Moran's <i>I</i>	Sd(<i>I</i>)	<i>p</i>
2010	0.413***	0.0082	0				
2011	0.408***	0.0082	0				
2012	0.403***	0.0082	0				
2013	0.398***	0.0082	0				
2014	0.396***	0.0082	0	2014	0.025	0.0104	0.0136
2015	0.392***	0.0082	0	2015	0.033**	0.0104	0.0011
2016	0.366***	0.0082	0	2016	0.072***	0.0105	0
2017	0.373***	0.0082	0	2017	0.148***	0.0105	0

Note: The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.